Democratising dialect: crowdsourcing language data across geographic space

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Dialectology has a long tradition of reaching out to the crowd to map language use across time and space (e.g. Ellis 1889). The 21 st century has seen a resurgence in such studies (e.g. Vaux and Bert 2013, Leemann et al. 2018) but in contrast to the face-to-face pen and paper questionnaires targeted at a socially restricted sample of speakers in traditional dialectology, these surveys utilise online technologies to reach out to the wider voices of the crowd. The sheer scale of these studies has obvious advantages in capturing the dialect landscape in all its forms, from rural to urban, young to old, and across a range of social classes. And beyond linguistic research, they have the potential to significantly democratise participation in, and access to, knowledge. However, given the relatively uncontrolled context in which data are collected, how reliable is crowdsourcing in documenting the complexities of dialect variation? As Brunt & Meidell (2017:56) point out 'If we are to rely on crowdsourcing to gather [...] information, then we need to be sure that the information is [...] accurate, and representative'. On the one hand, crowdsourced data is demonstrated to reflect patterns of use found in face-to-face, supervised research (e.g. Blaxter & Britain 2021), but a number of studies point to a range of possible issues with this method, including sampling and lack of quality control in such an unsupervised platform (e.g. Leemann et al. 2018, MacKenzie et al. 2022).

In this paper we present findings from a new crowdsourced resource - *Speak for Yersel* (speakforyersel.ac.uk) - which sets out to map dialect use in Scots throughout Scotland. Scotland is a small country but is said to pack a punch in terms of dialect diversity: how successful is crowdsourcing in revealing Scots in all its complex dialect guises? We first begin by describing the creation of *Speak for Yersel* and then address the questions of representativeness and accuracy. The paper ends with information on the expansion of *Speak for Yersel* and the open access tool we have made available for other researchers in the field.

The *Speak for Yersel* website aims to capture the different lexical, phonetic and morphosyntactic forms used across Scotland. This is an ambitious task, given the complexities of Scots, and even more difficult given the numerous constraints that crowdsourcing presents, including 1. providing a representative overview of forms used in Scots, 2. having questions that were both limited enough in number to avoid user fatigue and 3. simple enough that participants understood what was being asked (e.g. Blaxter & Britain 2021:1). The survey consisted of 87 questions in total, with responses automatically added to publicly accessible interactive maps on submission. In the first two weeks, during which the resource was widely publicised in both traditional and social media, we garnered c5k respondents, representing around 0.1% of the Scottish population.

With regards to representation, 'because crowds are not selected purposefully, they do not represent any particular population and tend to vary in terms of national representation' (Stewart et al. 2017: 737). Studies show that participation in online platforms may be skewed towards younger, educated, urban females (e.g. Vuorio & Horne 2023). Such user profiles may have significant repercussions for dialect studies where social attributes align closely with the dialect forms used. To assess representativeness in the current resource, we compared the data derived from the *Speak for Yersel* registration page on age, gender, geographic location and education to those reported in the 2022 Census for Scotland. For geography, age and gender, there is close mapping with the Census data, suggesting that the participant sample is representative of the population more broadly (see Figure 1). Education, however, is less well represented: 70% of our participants went to university, compared to a national average of c30%.

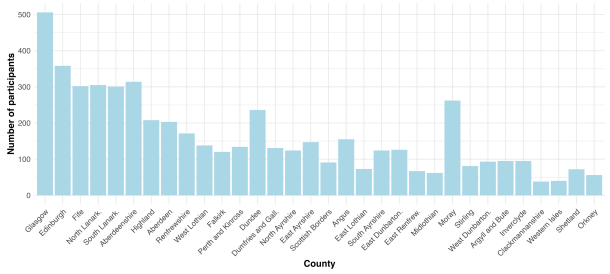

Figure 1a: Population of Scotland by region in the 2022 Census

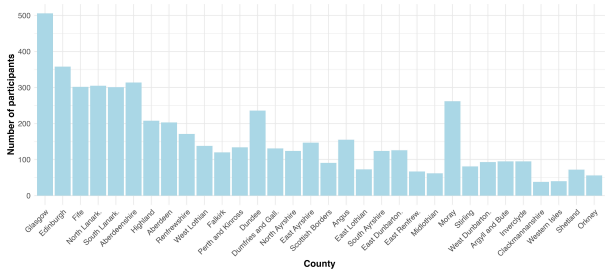

Figure 1b: Population of Scotland in Speak for Yersel

With regards to accuracy, the complexities of language use usually require highly controlled methods in collecting data. A key question is whether unsupervised crowdsourced data can be as reliable. We have an ideal opportunity to test this by comparing the *Speak for Yersel* data with data gathered using traditional face-to-face questionnaires for the *Scots Syntax Atlas* (scotssyntaxatlas.ac.uk). Using one linguistic form as an exemplar (*Where's the squirrel? It's there/there it is/ there it's there/there it's.*), we provide a comparison of results from both datasets. Even with this highly complex form which demonstrates geographic nuances of use, the results suggest that data elicited through crowdsourcing provides responses in line with those gathered in more controlled environments.

Given these positive results, we have now expanded to other areas: *Speak for Yersel* Wales, Northern Ireland and the Republic of Ireland successfully launched in October 2024. In order to provide the opportunity to archive language use even more widely, we have published the platform created for *Speak for Yersel* as a freely available open-source tool (github.com/baitken/sfy). In making this tool more widely available, we hope to further advance access and accessibility to knowledge of the linguistic landscape in Scotland, the UK and elsewhere.

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